

KI-HOON LEE

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Final-year Mechanical Engineering student with hands-on experience in hardware development, manufacturing, and system integration. Skilled in product design, quality control, and data analysis. My EAD starts in June 2025.

EDUCATION

UNIVERSITY OF CALIFORNIA, Berkeley Master of Engineering - <u>Mechanical Engineering - Advanced Energy Technology track</u> <i>Courses: Data Science, Energy Conversion, Machine Learning, Microprocessor-Based Mechanical Systems, Nanotechnology</i>	<u>Aug 2024- May 2025</u> Berkeley, CA
UNIVERSITY OF BATH Bachelor of Engineering - Mechanical Engineering	<u>Sep 2019- Jun 2024</u> Bath, UK

PROFESSIONAL EXPERIENCE

<u>Product Design Engineer</u> BLUE GOJI	<u>Sep 2024 – Present</u> Austin, TX
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- Designed and manufactured a resistive omnidirectional treadmill for a gamified fitness experience, that allows for movement without motor actuation that provides smooth braking for user control.
- Proposed and implemented a hybrid braking system combining friction and magnetic braking, reducing costs by 40% while maintaining safety, and refined the resistive mechanism for controlled movement through scaled-down prototyping.
- Conducted FEA simulation and iterative testing, to enhance performance, simplify manufacturing, and improve system efficiency.

<u>Reliability Engineering Intern</u> KIMM	<u>Aug - Nov 2020</u> Daejeon, Korea
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- Defined and executed design verification test plans for 6-axis robotic arm and cardan shaft hardware, conducting load testing, fatigue analysis, and vibration testing to ensure integrity, durability, and compliance with industry standards.
- Analyzed 500+ hours of lab and field test data, calculating failure rates, leading to design recommendations that improved component lifespan by 15%

<u>Materials Research Intern</u> KOREA UNIVERSITY	<u>Jul – Sep 2019</u> Seoul, Korea
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- Conducted material analysis using TEM to examine the microstructures of polymers and optimized nanoparticle dopants, achieving ~2% increase in infrared emittance and improving solar reflectance for the development of a passive cooling paint.
- Assisted with hydrogen energy projects by developing data analysis methods, processing experimental results, and identifying efficiency trends, leading to improved insights on hydrogen storage performance and reaction kinetics.

<u>GIS Software Intern</u> KOREA SPATIAL INFORMATION & COMMUNICATIONS	<u>Jul – Sep 2023</u> Seoul, Korea
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- Developed a live-updating informational map, emphasizing accuracy, quick real time updates and detailed info utilizing QGIS.
- Created multiple informational maps for the company website, focusing on functionality, aesthetics and geospatial precision.

<u>DMZ Infantry Squad Leader</u> REPUBLIC OF KOREA ARMY	<u>Jan 2021 – Jun 2022</u> Demilitarized Zone, Korean Border
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- Led and trained a squad of 8, as a sergeant and a squad leader in a high-stakes environment, one mile away from a North Korean GP, ensuring team readiness and operational effectiveness by running day and night shifts, patrols and war preparation trainings.

UNDERGRADUATE RESEARCH

<u>LALe Solar Powered UAV Optimization</u> LIMOSAERO	<u>Jan – May 2020</u> Cambridge, UK
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- Developed a flight simulation for Limosaero's solar-powered UAV, working with the company to evaluate performance across varying conditions of cloud coverage, tropical year, and velocity by implementing real-time UAV input and weather updates.
- Analyzed and optimized flight strategies to balance velocity, trajectory, and energy consumption, enhancing UAV performance and extending its operational range by ~50%.

LANGUAGE & SKILL

Languages: Korean: Fluent, French: DELF B2 (Upper Intermediate), Chinese: HSK 1 (Beginner)

Skills: SolidWorks, Autodesk (Inventor, Fusion 360, AutoCAD), FEA (Nastran, Ansys), 3D Printing, Manufacturing, Prototyping, Product Management, MATLAB, Python (TensorFlow, PyTorch, Pandas), Machine Learning, LabVIEW, QGIS, Microsoft Azure